

4 ISLR 4.8.7

Let class 1 = 'Yes'

Mean of class 1 i.e. $\mu_1 = 10$

Let class 0 = 'No'

Mean of class 0 i.e. $\mu_0 = 0$

Variance : $\sigma^2 = 36$

Prior Probabilities :

class 1 : $P(\text{Yes}) = 0.8$

class 0 : $P(\text{No}) = 0.2$

$x \rightarrow$ Normally distributed, observed $x=4$ for last year.

Applying Naive Bayes classifier and taking normal distribution of both classes.

$$f(x, \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\left(\frac{(x-\mu)^2}{2\sigma^2}\right)}$$

$$\begin{aligned} \text{For class 1 : } f(4) &= \frac{1}{\sqrt{2\pi \cdot 36}} e^{-\left(\frac{(4-10)^2}{2 \cdot 36}\right)} \\ &= \frac{1}{\sqrt{72\pi}} \cdot e^{-0.5} \end{aligned}$$

Using Bayes theorem:

$$P(\text{Yes} | x=4) = \frac{f_{\text{yes}}(4) \cdot P(\text{Yes})}{f_{\text{yes}}(4) \cdot P(\text{Yes}) + f_{\text{no}}(4) \cdot P(\text{No})}$$

For class 0%

$$f_{\text{class} = 0\%}(4) = \frac{1}{\sqrt{2\pi \cdot 36}} e^{-\frac{(4-0)^2}{2 \times 36}}$$
$$= \frac{1}{\sqrt{72\pi}} \cdot e^{-2/9}$$

$$\therefore P(\text{Yes} | x=4) = \frac{e^{-0.5} \cdot 0.8}{0.8 \cdot e^{-0.5} + e^{-2/9} \cdot 0.2}$$
$$= \frac{0.606 \times 0.8}{0.606 \times 0.8 + 0.8 \times 0.2}$$
$$= 0.7518 \cdot P$$

$$P(\text{Yes} | x=4) = 0.7518 = \boxed{75.18\%}$$

chance of dividend